

## Solution Requirements

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| <b>Date</b>          | 11 July 2026                                                          |
| <b>Team ID</b>       | SWTID-2026-8118                                                       |
| <b>Project Name</b>  | RK Health – AI Smart Patient Appointment & Medication Reminder System |
| <b>Maximum Marks</b> | 4 Marks                                                               |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

|   |                        |                                                                                                                                                      |      |
|---|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1 | Appointment Management | Users can create, view, search, filter, paginate, edit and delete patient appointments, with live status counts (Today/Confirmed/Pending/Cancelled). | High |
| 2 | Medication Management  | Users can add, edit, delete and track medications with dosage, frequency, start/end date and reminder status per patient.                            | High |
| 3 | Dashboard              | A central dashboard shows real-time counts for appointments, medications, AI summaries and reports.                                                  | High |
| 4 | Reports                | Users can generate, search, filter, export (CSV/PDF) and print patient reports.                                                                      | High |

|   |                                   |                                                                                                                       |        |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|
| 5 | AI Summary                        | System generates AI-based patient health summaries, risk levels and medication adherence insights.                    | Medium |
| 6 | External interfaces               | Frontend communicates with Google Apps Script (deployed as a web app) which reads/writes to Google Sheets.            | High   |
| 7 | Search, Filter & Pagination       | Every module (Appointments, Medications, Reports) supports keyword search, status/date filters and paginated results. | Medium |
| 8 | Data export                       | Reports and appointment/medication tables can be exported as CSV and printed directly from the browser.               | Medium |
| 7 | Compliance to laws or regulations | <i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i>         |        |
| 8 | <i>Other</i>                      | <i>[Add any additional functional requirements here]</i>                                                              |        |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|   |                            |                                                                                                                         |                                                                                 |
|---|----------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1 | Performance & Speed        | Dashboard and module pages should load quickly on GitHub Pages with Google Apps Script responses.                       | Dashboard loads in 2–3 seconds; CRUD actions respond in 1–2 seconds             |
| 2 | Scalability                | The system should comfortably handle the record volumes typical of a small clinic's appointment/medication logs.        | Handles hundreds of records per sheet without noticeable slowdown               |
| 3 | Security & Data Privacy    | Communication between frontend and Apps Script should avoid CORS/data-leak issues; only intended endpoints are exposed. | No CORS errors in production; requests limited to the deployed Apps Script URL  |
| 4 | Reliability & Availability | The app should remain available as a static GitHub Pages site with a stable Apps Script backend.                        | Consistent uptime in line with GitHub Pages and Google Apps Script availability |
| 5 | Usability & Accessibility  | Interface should be clean, responsive and usable on both desktop and mobile screens.                                    | Responsive layout verified on desktop and mobile viewports                      |
| 6 | Maintainability            | Codebase should remain modular across dashboard, appointments, medications, reports and AI summary pages.               | Each module kept in its own page/section for easy updates                       |